From Knowing to Doing: Learning Diverse Motor Skills through Instruction Learning

Linqi Ye, Jiayi Li, Yi Cheng, Xianhao Wang, Bin Liang, Yan Peng

***Abstract*— Recent years have witnessed many successful trials in the robot learning field. For contact-rich robotic tasks, it is challenging to learn coordinated motor skills by reinforcement learning. Imitation learning solves this problem by using a mimic reward to encourage the robot to track a given reference trajectory. However, imitation learning is not so efficient and may constrain the learned motion. In this paper, we propose instruction learning, which is inspired by the human learning process and is highly efficient, flexible, and versatile for robot motion learning. Instead of using a reference signal in the reward, instruction learning applies a reference signal directly as a feedforward action, and it is combined with a feedback action learned by reinforcement learning to control the robot. Besides, we propose the action bounding technique and remove the mimic reward, which is shown to be crucial for efficient and flexible learning. We compare the performance of instruction learning with imitation learning, indicating that instruction learning can greatly speed up the training process and guarantee learning the desired motion correctly. The effectiveness of instruction learning is validated through a bunch of motion learning examples for a biped robot and a quadruped robot, where skills can be learned typically within several million steps. Besides, we also conduct sim-to-real transfer and online learning experiments on a real quadruped robot. Instruction learning has shown great merits and potential, making it a promising alternative for imitation learning.**

***Index Terms*—Reinforcement learning, legged locomotion, instruction learning, quadruped robots, biped robots.**

I. Introduction

With the successful application to legged robots [1][2], aerial robots [3], and robotic manipulation [4], the learning method has been proven to be an efficient way for robot motion control. Reinforcement learning aims to maximize a reward through interactions with the environment and observations of how it responds. Although many powerful reinforcement learning methods have been proposed, such as deep deterministic policy gradient (DDPG) [5], proximal policy optimization (PPO) [6], and soft actor critic (SAC) [7], the reward design remains a challenging problem, especially for contact-rich tasks. For example, when learning legged locomotion, a simple reward may result in unnatural motion behaviors [8]. Although the robot can learn to move fast and transverse obstacles, it behaves awkwardly.

Imitation learning provides a way to generate natural-looking behaviors by applying a motion reference signal to the reward, which is usually obtained from motion capture data [9]. However, imitation learning does have some drawbacks. Firstly, the imitation process is reward-driven, where the robot starts from random actions and tries to improve it to obtain a higher reward. This process is different from the human learning process. For example, when learning to dance, we can do it the first time according to the teacher's demonstration or instruction, just not so well. Then we do it better and better with more trials, just as the saying says "practice makes perfect". In this way, we can learn sophisticated skills very efficiently. However, imitation learning usually has very bad starts and takes millions of trials to finally master the desired motion. Secondly, imitation learning tries to minimize the error between the robot's motion and the reference motion, which constrains the robot and may prevent it from exploring behaviors that have better performance. Whereas for human learning skills, we are not trying to exactly copy the behavior of others, but take it as a good start and try to make it best suit ourselves.

Inspired by the motion skill learning process of humans, we propose the instruction learning framework. The core of instruction learning is "knowing-to-doing", where the robot starts from an instructed action and improves it through training. Compared to imitation learning, which uses a reference signal in the reward, instruction learning uses the reference signal directly in the action as feedforward and applies reinforcement learning to make corrections to achieve a given goal. The learning mechanism makes instruction learning inherently more efficient and less constrained.

The contributions of this paper are fourfold. First, we propose the concept of instruction learning and elaborate on its similarities with human learning and essential differences with imitation learning. Although [10] has proposed a similar feedforward-feedback learning structure, instruction learning is different in two aspects: 1) We propose the action bounding technique by using discounted feedback, which is the key for fast learning; 2) We remove the mimic reward in instruction learning, which gives it more flexibility. By using instruction learning, the robots can learn various locomotion behaviors with a simple stepping motion as feedforward. Second, we fully demonstrate the efficiency, versatility, and flexibility of instruction learning through a bunch of locomotion learning examples, including bipedal walk, level walk, march walk, jump, and hop, as well as quadruped trot, pace, bound, and pronk. Third, we investigate another technique that is neglected in previous studies of imitation learning, that is,

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The code is available at https://github.com/Lr-2002/raisimLib.

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从认知到实践：通过指令学习掌握多样运动技能

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摘要— 近年来，机器人学习领域取得了许多成功的尝试。对于接触丰富的机器人任务，通过强化学习来学习协调运动技能具有挑战性。模仿学习通过使用模仿奖励来鼓励机器人跟踪给定的参考轨迹，从而解决了这一问题。然而，模仿学习效率不高，且可能限制所学到的运动。在本文中，我们提出了指令学习，该方法受人类学习过程启发，具有高效、灵活且通用的特点，适用于机器人运动学习。指令学习不将参考信号用于奖励，而是直接将参考信号作为前馈动作应用，并与通过强化学习学到的反馈动作相结合来控制机器人。此外，我们提出了动作边界技术并移除了模仿奖励，这对于高效和灵活的学习至关重要。我们比较了指令学习与模仿学习的性能，表明指令学习可以极大地加速训练过程，并确保正确学习期望的运动。通过针对双足机器人和四足机器人的一系列运动学习示例，验证了指令学习的有效性，其中技能通常可以在几百万步内学会。此外，我们还在真实四足机器人上进行了仿真到现实迁移和在线学习实验。指令学习展现了巨大的优势和潜力，使其成为模仿学习的一个有前景的替代方案。

索引词—强化学习，腿部运动，指令学习，四足机器人，双足机器人。

I. 引言

随着在腿式机器人 [1][2],、空中机器人 [3], 和机器人操作 [4], 中的成功应用，学习方法已被证明是机器人运动控制的有效途径。强化学习旨在通过与环境的交互以及对其响应的观测来最大化奖励。尽管已经提出了许多强大的强化学习方法，例如深度确定性策略梯度（DDPG） [5], 、近端策略优化（PPO）[6], 和软演员评论家（SAC）[7],，但奖励设计仍然是一个具有挑战性的问题，尤其是在接触密集的任务中。例如，在学习腿部运动时，简单的奖励可能导致不自然的运动行为[8]。尽管机器人可以学会快速移动和穿越障碍物，但其行为显得笨拙。

模仿学习通过将运动参考信号应用于奖励（通常从动作捕捉数据中获取）来生成自然的行为 [9]。然而，模仿学习确实存在一些缺点。首先，模仿过程是奖励驱动的，机器人从随机动作开始，并试图改进以获得更高的奖励。这个过程与人类学习过程不同。例如，在学习跳舞时，我们可以根据老师的示范或指导第一次就做出动作，只是做得不够好。然后通过更多的尝试，我们做得越来越好，正如俗话所说"熟能生巧"。通过这种方式，我们可以非常高效地学习复杂的技能。然而，模仿学习通常起步非常糟糕，需要数百万次尝试才能最终掌握所需的动作。其次，模仿学习试图最小化机器人动作与参考动作之间的误差，这限制了机器人，并可能阻止其探索性能更优的行为。而对于人类学习技能而言，我们并非试图完全复制他人的行为，而是将其作为一个良好的起点，并努力使其最适合自己。

受人类运动技能学习过程的启发，我们提出了指令学习框架。指令学习的核心是"知行合一"，即机器人从指令动作出发，通过训练进行改进。与在奖励中使用参考信号的模仿学习不同，指令学习将参考信号直接作为前馈应用于动作中，并运用强化学习进行修正以实现给定目标。这种学习机制使得指令学习本质上更高效且约束更少。

本文的贡献有四个方面。首先，我们提出了指令学习概念，并详细阐述了其与人类学习的相似性以及与模仿学习的本质区别。尽管 [10] 已经提出了类似的前馈-反馈学习结构，但指令学习在两个方面有所不同：1) 我们提出了使用折扣反馈的动作边界技术，这是快速学习的关键；2) 我们在指令学习中移除了模仿奖励，这赋予了它更大的灵活性。通过使用指令学习，机器人能够以简单的踏步运动作为前馈，学习各种运动行为。其次，我们通过一系列运动学习实例充分展示了指令学习的效率、通用性和灵活性，包括双足行走、平地行走、正步走、跳跃和单脚跳，以及四足小跑、踱步、弹跳和蹦跳。第三，我们研究了在以往模仿学习研究中被忽视的另一项技术，即：

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reference in observation (RO). In most studies of imitation learning, the reference signal is put in the reward but not added to the observation, which we think is one root that causes its inefficiency. We show that RO can facilitate imitation learning to learn faster and correctly. However, even with RO, imitation learning is still not as efficient as instruction learning, which might be because of their inherent learning mechanism. Last, we demonstrate the feasibility of instruction learning on real systems by taking sim-to-real transfer and online learning experiments on a real quadruped robot. Thanks to the high efficiency of instruction learning, the robot can learn in the real world to walk forward or turn around in only 20 minutes.

II. Related Work

A. *Robot Reinforcement Learning*

Reinforcement learning has become one of the mainstreams for robot locomotion control, which has led to many remarkable achievements in recent years. Duan et al. [11] propose an approach that combines bipedal robot control with reinforcement learning to allow for feet setpoint learning in task space and develop a reinforcement learning formulation for training dynamic gait controllers that can respond to specified touchdown locations. Siekmann et al. [12] index time via a cycle time variable with a periodic reward function accordingly, and common bipedal gaits including walking, running, hopping, galloping, and skipping are realized. Learning-based state estimation networks and dynamics randomization are introduced to better demonstrate fast and robust locomotion on rough terrains [13]. Margolis et al. [14] use a teacher-student training policy and a few adaptive curriculum update rules to realize fast moving in both indoor and outdoor running. Hwangbo et al. [15] leverage an attention-based recurrent encoder that integrates proprioceptive and exteroceptive input. They successfully deployed the controller on the ANYmal C robot which traversed challenging natural environments with steep inclination, slippery surfaces, grass, and snow. Margolis et al. [16] propose learning a single policy that encodes a structured family of locomotion strategies that solve training tasks in different ways evolving a conditional part to specify tasks, to avoid the slow and iterative cycle of reward and environment redesign.

Great progress based on reinforcement learning has been achieved so far, and some new frameworks are explored to accelerate the learning process. Kim et al. [17] find that leveraging constraints in the learning pipeline is promising to replace the substantial and laborious reward engineering and propose a novel framework for training neural network controllers for complex robotic systems consisting of both rewards and constraints. A reinforcement learning framework using massive parallelism on a single GPU [18] was proposed and the training time can be reduced by multiple orders of magnitude compared to previous work. It is demonstrated that a complex real-world robotics task can be trained in minutes using NVIDIA's Isaac Gym simulation environment [19].

B. *Robot Imitation Learning*

For robot reinforcement learning, reward design is a crucial and challenging part. It's more intuitive for controllers like trajectory optimization methods to generate complex movements such as backflips for legged robots, while designing rewards for reinforcement learning methods to achieve this is difficult. Imitation learning is a good way to pick up pattern priori from the experts' experiences as a simple reward term, as these experiences may not be intuitive enough to form mathematical expressions to generate different motion characteristics like maintaining adequate foot clearances and behaving naturally. To some extent, imitation learning leverages expert knowledge and avoids complex reward design.

The easiest way to achieve imitation learning is to use a mimic reward to encourage the robot to follow a given motion. The mimic reward is the error between the robot state and the reference trajectory to be followed. In [9], the captured motion data from a real dog is used as reference trajectories for a quadruped robot to implement imitation learning, which successfully learns a diverse set of natural behaviors. However, imitation learning usually has low efficiency, sometimes it learns nothing. Therefore, the reference state initialization and early termination techniques are proposed in [20], which are found to be critical for achieving highly dynamic skills learning. Besides, early termination is further extended to spacetime bounds in [10], where the episodes are terminated immediately once any spacetime bounds are violated. Another problem with imitation learning is that the resulting motion is constrained to the given reference. To solve this problem, a task reward is introduced in [20] to enable adjustment of the learned behavior. In [21], a learning curriculum with imitation relaxation is designed, which can eliminate the influence of the non-ideal reference trajectory and excavate the full potential of the robot.

Behavioral cloning (BC), inverse reinforcement learning (IRL), and generative adversarial imitation learning (GAIL) also belong to imitation learning [22]. BC [23] learns a policy as a supervised learning problem and generates the state-action trajectory by matching the teaching movements. IRL [24] seeks to best explain the observed behavior by learning the corresponding reward function, and the policy is learned through an iterative loop of the Markov decision process solving and reward function updating. GAIL [25] applies the ideas of GANs [26][27] to imitate an expert. It directly extracts a policy from data and avoids significant computational expense in learning the reward function. Baram et al. [28] introduce the Model-based Generative Adversarial Imitation Learning (MGAIL) algorithm which uses relatively fewer expert samples and interactions with the environment. A forward model is given to make the computation fully differentiable, which enables training policies using the exact gradient of the discriminator. Merel et al. [29] extend GAIL to enable training for generic neural network policies and produce humanlike movement patterns from limited demonstrations. Peng et al. [30][31] propose adversarial motion priors (AMP), which combine adversarial imitation learning and unsupervised learning to develop skill embeddings that produce life-like behaviors. Task-specific annotation or segmentation of the motion data is needless and models can be trained using large datasets of unstructured motion clips. Vollenweider et al. [32] introduce multi-AMP which augments the concept of AMP to allow for multiple, discretely switchable styles. Their approach is validated in real-world experiments with a wheeled-legged quadruped

观测中的参考信号 (RO)。在大多数模仿学习研究中，参考信号被置于奖励中而非添加到观测中，我们认为这是导致其效率低下的根源之一。我们证明，RO 能够促进模仿学习更快、更准确地学习。然而，即使采用 RO，模仿学习的效率仍然不如指令学习，这可能是由于其固有的学习机制所致。最后，我们通过在真实四足机器人上进行仿真到现实迁移和在线学习实验，展示了指令学习在真实系统上的可行性。得益于指令学习的高效率，机器人只需 20 分钟即可在现实世界中学会向前行走或转弯。

II. 相关工作

A. 机器人强化学习 强化学习已

成为机器人运动控制的主流方法之一，近年来取得了许多显著成就。Duan 等人 [11] 提出了一种将双足机器人控制与强化学习相结合的方法，允许在任务空间中进行足部设定点学习，并开发了一种强化学习公式，用于训练能够响应指定触地位置的动态步态控制器。Siekmann 等人 [12] 通过周期时间变量和相应的周期性奖励函数对时间进行索引，实现了常见的双足步态，包括行走、跑步、单脚跳、疾驰和跳跃。引入基于学习的状态估计网络和动力学随机化，以更好地在崎岖地形上展示快速且鲁棒的运动 [13]。Margolis 等人 [14] 使用师生训练策略和少量自适应课程更新规则，实现了室内外跑步中的快速移动。Hwangbo 等人 [15] 利用基于注意力的循环编码器，整合了本体感觉和外部感觉输入。他们成功地将控制器部署在 ANYmal C 机器人上，该机器人穿越了具有陡峭坡度、光滑表面、草地和雪地等挑战性的自然环境。Margolis 等人 [16] 提出学习一种单一策略，该策略编码了一个结构化的运动策略族，以不同方式解决训练任务，并演化出一个条件部分来指定任务，从而避免奖励和环境重新设计的缓慢迭代循环。

基于强化学习的重大进展迄今已取得，一些新框架被探索以加速学习过程。Kim等人 [17] 发现，在学习流程中利用约束有望取代繁重且费力的奖励工程，并提出了一种新颖框架，用于训练由奖励和约束共同构成的复杂机器人系统的神经网络控制器。一种利用单GPU大规模并行化的强化学习框架[18] 被提出，其训练时间相比先前工作可减少多个数量级。研究表明，使用NVIDIA Isaac Gym 仿真环境[19]，复杂的现实世界机器人任务可在数分钟内完成训练。

B. 机器人模仿学习

对于机器人强化学习而言，奖励设计是一个关键且具有挑战性的部分。对于像轨迹优化方法这样的控制器来说，生成复杂的动作更为直观，例如腿式机器人的后空翻，而设计用于强化学习方法的奖励来实现这一目标则十分困难。模仿学习是一种从专家经验中获取先验模式作为简单奖励项的有效方式，因为这些经验可能不足以直观地形成数学表达式来生成不同的运动特征，例如保持足够的足部离地间隙和自然的行为。在某种程度上，模仿学习利用了专家知识，并避免了复杂的奖励设计。

实现模仿学习最简单的方法是使用模仿奖励来鼓励机器人跟随给定动作。模仿奖励是机器人状态与待跟随参考轨迹之间的误差。在[9], 中，从真实狗身上捕捉的运动数据被用作四足机器人的参考轨迹以实现模仿学习，成功学习了一系列多样化的自然行为。然而，模仿学习通常效率低下，有时甚至学不到任何东西。因此，参考状态初始化和提前终止技术在[20], 中被提出，这些技术被发现对实现高动态技能学习至关重要。此外，在[10], 中，提前终止被进一步扩展为时空边界，一旦任何时空边界被违反，回合立即终止。模仿学习的另一个问题是，生成的运动受限于给定的参考。为解决此问题，在[20] 中引入了任务奖励，以允许对学习行为进行调整。在[21], 中，设计了一种带有模仿松弛的学习课程，可以消除非理想参考轨迹的影响，并挖掘机器人的全部潜力。

行为克隆（BC）、逆强化学习（IRL）和生成对抗模仿学习（GAIL）也属于模仿学习[22]。BC[23] 将策略学习视为一个监督学习问题，通过匹配教学动作来生成状态-动作轨迹。IRL [24]旨在通过学习相应的奖励函数来最佳地解释观察到的行为，并通过马尔可夫决策过程求解与奖励函数更新的迭代循环来学习策略。GAIL[25] 应用生成对抗网络[26][27] 的思想来模仿专家。它直接从数据中提取策略，避免了学习奖励函数时的大量计算开销。Baram等人 [28]提出了基于模型的生成对抗模仿学习（MGAIL）算法，该算法使用相对较少的专家样本和与环境交互的次数。通过引入前向模型使计算完全可微，从而能够利用判别器的精确梯度来训练策略。Merel等人[29] 扩展了GAIL，使其能够训练通用的神经网络策略，并从有限的示范中生成类人运动模式。Peng等人 [30][31] 提出了对抗运动先验（AMP），该方法结合了对抗模仿学习和无监督学习，以开发能够产生逼真行为的技能嵌入。无需对运动数据进行特定任务的标注或分割，模型可以使用包含非结构化运动片段的大型数据集进行训练。Vollenweider等人[32] 引入了 multi-AMP，它扩展了AMP的概念，允许实现多种可离散切换的风格。他们的方法在轮腿式四足机器人的实际实验中得到了验证

robot switching between a quadrupedal and a humanoid configuration.

C. Robot Online Learning

In a standard paradigm of robot reinforcement learning, the controller is conventionally trained first in the simulation environment and then transferred to the real robot. The sim-to-real gap which derives from the model mismatch, unpredictable environmental changes, controller delay, and other factors makes it a challenge for algorithms to work in real systems. Therefore, various techniques are proposed to solve this problem.

One solution is to increase the robustness of the policy in simulation before the algorithm deployment. Domain randomization is introduced to bridge the "reality gap" by training a policy on the distribution of environmental conditions in simulation [33]-[35]. Kumar et al. [36] present the rapid motor adaptation algorithm including an adaptation module as a subsystem to estimate the environment configuration using the information on the robot's recent state and action history. Zhuang et al. [37] distill their policy before deployment. They apply pre-processing techniques to both the raw rendered depth image and the raw real-world depth image to bridge the sim-to-real gap in visual appearance. Tan et al. [38] introduce a hierarchical framework that uses a high-level learned policy to adjust the low-level trajectory generator for better adaptability to the terrain. However, these algorithms generate relatively conservative behavior for the policy to fit both simulation and real-world environments.

Another solution is applying online learning, namely training in the real world directly. Although online learning has been successfully verified on fixed-base robots such as robotic arms through a large amount of data training [39][40], researchers emphasize the prohibitively expensive cost of tunning on some floating-base physical systems which can be easily damaged through extensive trial-and-error learning so that sample efficiency and security issue is crucial for legged robots [41]. Ha et al. [42] overcome the safety challenge by developing a safety-constrained RL framework, where the pitch and roll angles of the robot's torso are limited. Besides, robots reset after each episode is another important issue for online learning. Yang et al. [43] adopt model predictive control to account for planning latency and embed prior knowledge of leg trajectories into the action space and yield an approach that requires only 4.5 minutes of real-world data collection to learn robust and fast gaits for a quadruped robot. Bloesch et al. [44] train a small humanoid robot to walk and interact with the environment relying only on onboard sensors and limited prior knowledge of the hardware. If the robot has fallen at the end of an episode, a robust pre-programmed feedforward controller to stand up will be woken up. Gupta et al. [45] propose a reset-free method by learning multiple tasks together since different combinations of tasks can serve to perform resets for each other. This technique avoids frequent human intervention. Smith et al. [46][47] present a system that enables online locomotion policy fine-tuning via a combination of autonomous data collection and reinforcement learning.

III. Concept of Instruction Learning

A. Instruction Learning Framework

The framework of instruction learning is shown in Fig. 1. Unlike traditional reinforcement learning where the control action is entirely determined by a neural network, the control action of instruction learning consists of two parts: a feedback part learned by a neural network and a feedforward part which follows an instructed action.

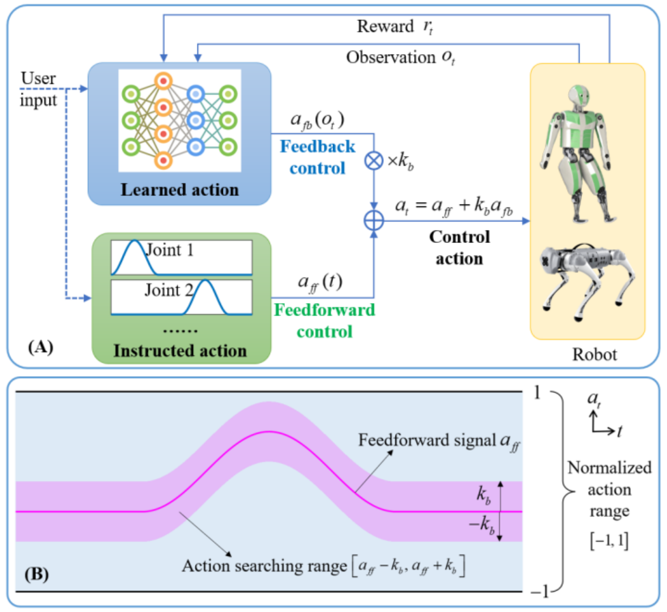


Figure 1. Instruction learning framework. (A) The feedforward-feedback control structure of instruction learning. (B) Illustration of action bounding.

The feedforward is joint angle trajectories concerning time, which is a priori knowledge. The feedback is a neural network that maps the robot's state to action, which is learned through reinforcement learning. The feedforward gives a rough routine and initial guess for the control action, while the feedback shapes it according to the reward to achieve a given goal such as maintaining balance and moving forward.

Since the control action is a combination of the feedforward and the feedback, the influence of the two components depends on their magnitudes. If the magnitude of the feedback is much larger than that of the feedforward, then the action is dominated by the feedback and thus the instruction role of the feedforward is weakened. Therefore, we introduce the action bounding technique, that is, multiplying the feedback by a ratio k_b before adding it to the feedforward. The feedback ratio determines the action bounding range, which constrains the magnitude of the feedback. By adjusting the feedback ratio, the boundaries of the action are specified so we can determine how far the action can be deviated from the feedforward action, as illustrated in Fig. 1 (B). We found that the feedback ratio is crucial for the learning efficiency of instruction learning. With the action bounding technique, the searching range of the action space is greatly reduced and thus can significantly accelerate the training process.

，该机器人可在四足与类人构型之间切换。

C. 机器人在线学习

在机器人强化学习的标准范式中，控制器通常先在仿真环境中训练，然后迁移到真实机器人上。源于模型失配、不可预测的环境变化、控制器延迟等因素的仿真到现实差距，使得算法在真实系统中工作面临挑战。因此，研究者提出了多种技术来解决这一问题。

一种解决方案是在算法部署前增强策略在仿真中的鲁棒性。通过引入域随机化，在仿真中环境条件的分布上训练策略，以弥合"现实差距"[33]-[35]。Kumar等人[36]提出了快速运动自适应算法，该算法包含一个自适应模块作为子系统，利用机器人近期状态和动作历史的信息来估计环境配置。Zhuang等人[37]在部署前对策略进行蒸馏。他们对原始渲染深度图像和原始真实世界深度图像均应用预处理技术，以弥合视觉外观上的仿真到现实差距。Tan等人[38]引入了一种分层框架，使用高层学习策略调整底层轨迹生成器，以更好地适应地形。然而，这些算法会生成相对保守的行为，以使策略同时适应仿真和真实世界环境。

另一种解决方案是应用在线学习，即直接在现实世界中进行训练。尽管在线学习已通过在机械臂等固定基座机器人上的大量数据训练 [39][40], 得到成功验证，但研究人员强调，在一些浮动基座物理系统上进行调优的成本极其高昂，这些系统容易因大量的试错学习而损坏，因此样本效率和安全性问题对于腿式机器人至关重要 [41]。Ha 等人 [42] 通过开发一个安全约束强化学习框架来克服安全挑战，该框架限制了机器人躯干的俯仰角和横滚角。此外，每个回合后机器人的重置是在线学习的另一个重要问题。Yang 等人 [43] 采用模型预测控制来处理规划延迟，并将腿部轨迹的先验知识嵌入到动作空间中，从而提出了一种方法，该方法仅需 4.5 分钟的真实世界数据收集即可为四足机器人学习稳健且快速的步态。Bloesch 等人 [44] 训练一个小型类人机器人仅依靠机载传感器和有限的硬件先验知识来行走并与环境交互。如果机器人在一个回合结束时摔倒，一个稳健的预编程前馈控制器将被唤醒以使其站立。Gupta 等人 [45] 提出了一种免重置方法，通过同时学习多个任务，因为不同的任务组合可以相互执行重置。这种技术避免了频繁的人工干预。Smith 等人 [46][47] 提出了一种系统，通过结合自主数据收集和强化学习来实现在线运动策略微调。

III. 指令学习概念

A. 指令学习框架 指令学习框架

学习过程如图 1 所示。与传统强化学习中控制动作完全由神经网络决定不同，指令学习的控制动作由两部分组成：一个由神经网络学习的反馈部分，以及一个遵循指令动作的前馈部分。

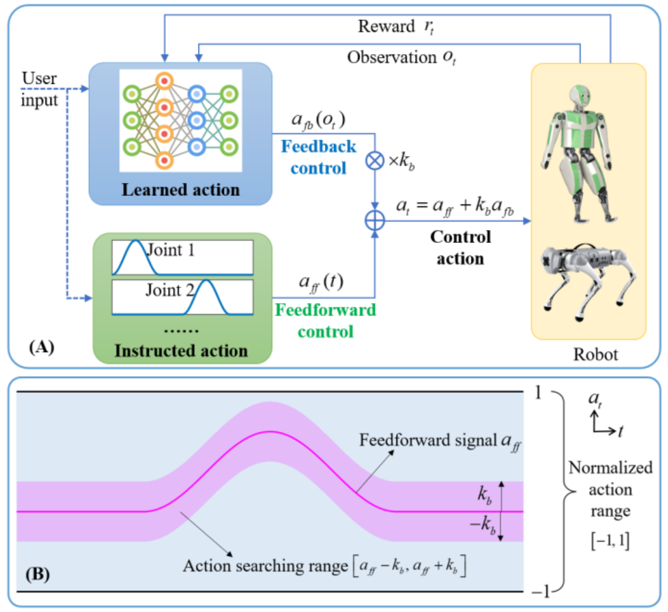


图1. 指令学习框架。(A) 指令学习的前馈-反馈控制结构。(B) 动作边界示意图。

前馈部分是关于时间的关节角度轨迹，属于先验知识。反馈部分是一个神经网络，它将机器人的状态映射为动作，并通过强化学习进行训练。前馈提供了控制动作的粗略路线和初始猜测，

而反馈则根据奖励对其进行调整，以实现诸如保持平衡和向前移动等给定目标。

由于控制动作是前馈与反馈的结合，两者的影响取决于其幅度。若反馈的幅度远大于前馈，则动作由反馈主导，从而削弱了前馈的指令作用。因此，我们引入了动作边界技术，即在将反馈与前馈相加之前，先将其乘以一个比率 。反馈比率决定了动作边界范围，该范围约束了反馈的幅度。通过k_b调整反馈比率，可以指定动作的边界，从而确定动作偏离前馈动作的程度，如图1(B)所示。我们发现，反馈比率对指令学习的学习效率至关重要。借助动作边界技术，动作空间的搜索范围大幅缩小，从而能显著加速训练过程。

B. *Association with Human Learning*

Fig. 2 shows the instruction learning process, which is inspired by humans. One example is learning to swim, the beginner has a prior knowledge of how to move the limbs in sequence but does not know how to coordinate the movements according to the body state in the water. The prior knowledge can be seen as feedforward and the coordination is feedback that needs to be learned from interaction with the water. Another example is the babies learning to walk. It is known that babies can take steps when held upright with the feet touching a solid surface, which is called the "stepping reflex" [48]. The stepping reflex is an important part of babies' development, which prepares them to take real steps several months down the road. The stepping reflex can also be viewed as feedforward, which is prior knowledge inherited from parents and works along with postnatal training to help babies walk.

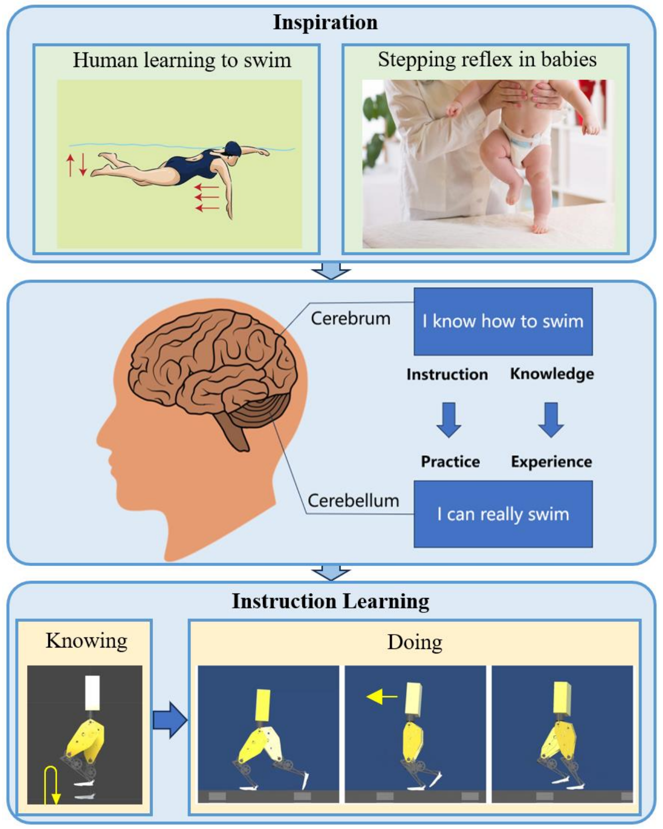


Figure 2. Association of instruction learning with human learning.

The counterparts of the feedforward and the feedback are knowledge and experience in the human brain, which are related to the cerebrum and cerebellum, respectively. For various human motion skills like swimming, dancing, bicycling, skiing, etc., we usually know of it first (usually from instruction), and then become experienced in it through practice. Similarly, the process of instruction learning is also from knowing to doing. For example, for a biped robot learning to walk, we can use the stepping motion as a feedforward. However, that is not enough for walking, the robot will fall if we put it on the ground. A feedback action is needed to stabilize the motion, and this is exactly what the reinforcement learning part is doing. Through massive trials and interactions, the robot learns to maintain balance, it can step in place, and can also walk at a given speed. In this learning process, the feedforward is very important for the robot to converge to a desired walking style.

Compared to the other reinforcement learning methods, instruction learning introduces an additional feedforward in the controller. Therefore, we now have two degrees of freedom during control design, that is, the reward and the feedforward. The feedforward directly affects the controller, while the reward gradually shapes the controller through training. Tunning the feedforward and reward helps us achieve different goals. For example, the feedforward can achieve periodic motion, a predefined motion sequence, or a given pose easily, while the reward can help maintain balance and motion coordination.

C. *Comparison with Imitation Learning*

Instruction learning and imitation learning both aim at reproducing some given motion styles, but their mechanisms are indeed very different. To illustrate this, we made a vivid comparison in Fig. 3.

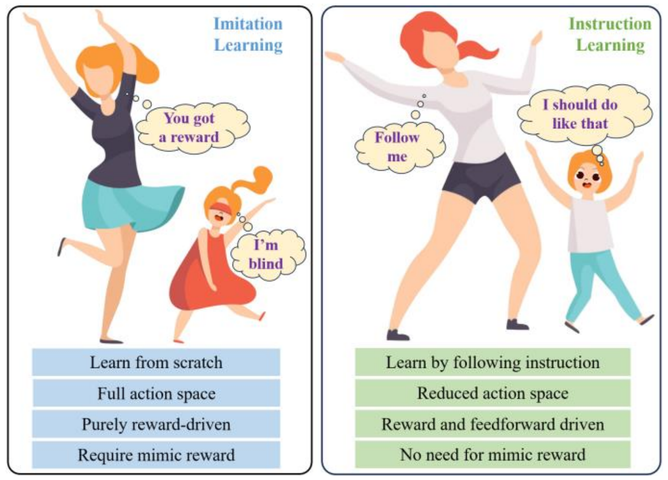


Figure 3. Imitation learning vs. instruction learning.

Imitation learning is like blind people learning to dance, the learner does not know the dancing movements in mind (the observation). It learns from scratch, explores in a full action space and the action is purely reward-driven. For instruction learning, the learner builds the concept of dancing by watching the teacher's movement, and then it learns by following instructions/demonstrations. It explores in a reduced action space around the instructed/demonstrated action. Its action is driven by both reward and feedforward. Therefore, the learning mechanism natively determines that instruction learning can be more efficient than imitation learning.

IV. Instruction Learning for Legged Locomotion

To verify the effectiveness of instruction learning, we take a biped robot and a quadruped robot as examples and apply instruction learning to learn different locomotion behaviors.

A. *Robot Model*

The simulated robot models are shown in Fig. 4. For the biped robot, we use a modified model of Ranger Max (previously called Cornell Tik-Tok [49]). The real Ranger Max

B. 与人类学习的关联

图2展示了受人类启发的指令学习过程。一个例子是学习游泳：初学者具备如何按顺序移动四肢的先验知识，但不知道如何根据水中身体状态协调动作。先验知识可视为前馈，而协调则是需要通过与水的互动来学习的反馈。另一个例子是婴儿学习行走。众所周知，当婴儿被竖直抱起且双脚接触坚实表面时，他们能迈出步伐，这被称为"踏步反射"[48]。踏步反射是婴儿发育的重要部分，为几个月后真正迈步做好准备。踏步反射也可视为前馈，即从父母遗传的先验知识，它与后天训练共同作用，帮助婴儿行走。

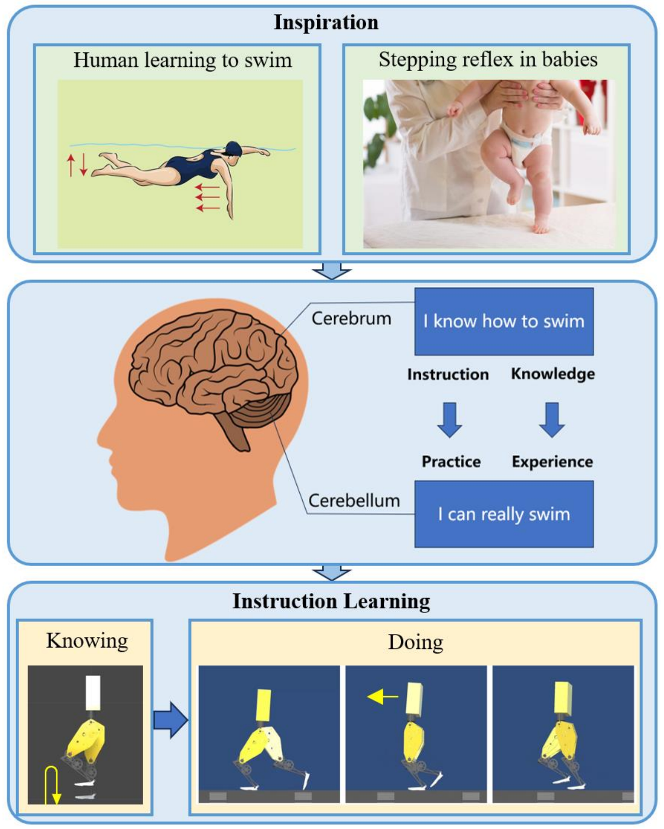


图2. 指令学习与人类学习的关联。

前馈与反馈在人类大脑中的对应物分别是知识与经验，二者分别与大脑和小脑相关。对于游泳、跳舞、骑自行车、滑雪等各种人类运动技能，我们通常先了解其原理（通常来自指令），然后通过实践积累经验。同样，指令学习的过程也是从认知到实践的过程。例如，对于学习行走的双足机器人，我们可以将踏步运动作为前馈。然而，仅凭这一点并不足以实现行走——如果将机器人放在地面上，它会摔倒。此时需要反馈动作来稳定运动，而这正是强化学习部分所承担的任务。通过大量的试验与交互，机器人学会了保持平衡，它能够原地踏步，还能以指定速度行走。在这一学习过程中，前馈对于机器人收敛至期望的行走姿态至关重要。

与其他强化学习方法相比，指令学习在控制器中引入了一个额外的前馈。因此，我们现在在控制设计中有两个自由度，即奖励和前馈。前馈直接影响控制器，而奖励则通过训练逐步塑造控制器。调整前馈和奖励有助于我们实现不同的目标。例如，前馈可以轻松实现周期性运动、预定义的运动序列或给定的姿态，而奖励则有助于维持平衡和运动协调。

C. 与模仿学习的比较

指令学习和模仿学习都旨在复现某些给定的运动风格，但它们的机制确实非常不同。为了说明这一点，我们在图3中进行了生动的比较。

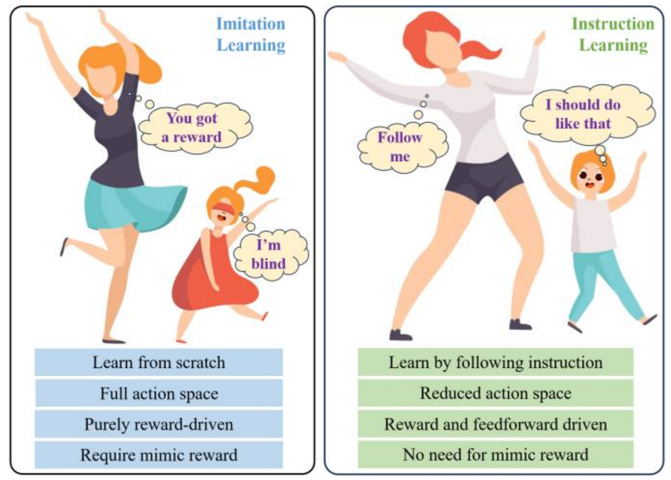


图3. 模仿学习与指令学习的对比。

模仿学习就像盲人学跳舞，学习者心中没有舞蹈动作的概念（即观测）。它从零开始学习，在完整的动作空间中探索，其动作纯粹由奖励驱动。而指令学习则是学习者通过观察老师的动作建立舞蹈概念，然后跟随指令/示范进行学习。它在指令/示范动作周围的缩减动作空间内探索，其动作由奖励和前馈共同驱动。因此，学习机制本身决定了指令学习可以比模仿学习更高效。

IV. 腿部运动的指令学习

为了验证指令学习的有效性，我们以双足机器人和四足机器人为例，应用指令学习来学习不同的运动行为。

A. 机器人模型

仿真机器人模型如图4所示。对于双足机器人，我们使用了Ranger Max（之前称为Cornell Tik-Tok [49]）的改进模型。真实的Ranger Max only has four joints in each leg. To enhance its capability, we add the hip yaw and ankle roll joints to the simulation model. For the quadruped robot, we use the Unitree A1 model. Specifications for the two robots are as follows.

Biped robot: Each leg has 6 actuated joints. The range of motion (take the position in Fig. 4 as zero) for each joint is (unit: degree): hip yaw -30~30, hip roll -30~30, hip pitch -60~120, knee pitch -170~10, ankle roll -30~30, ankle pitch -60~60. The maximum joint force is 200 Nm. The joint position control loop has a stiffness of 2000 and a damping of 100.

Quadruped robot: Each leg has 3 actuated joints. The range of motion (take the position in Fig. 4 as zero) for each joint is (unit: degree): hip roll -46~46, hip pitch -90~210, knee pitch -94.5~7.5. The maximum joint force is 33.5 Nm. The joint position control loop has a stiffness of 180 and a damping of 8.

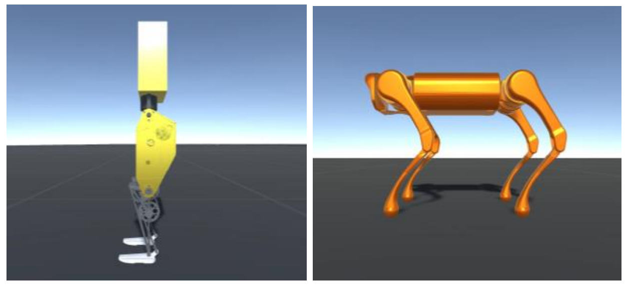


Figure 4. The simulated robot models. Left: Ranger Max; right: Unitree A1.

B. *Feedforward Design*

The feedforward is the position reference trajectory for each joint. For simplicity, we consider two commonly used trajectories.

The first is a periodic trajectory, where the joint moves cyclically with a fixed period. In this case, the reference angle is designed as a sinusoidal signal

$$\theta_t^{ref} = \theta_0 + \Delta\theta \frac{1-\cos\left(2\pi t / T\right)}{2} \tag{1}$$

The second is a point-to-point trajectory, where the joint moves from one position to another position. In this case, the reference angle is designed as a ramp signal

$$\theta_t^{ref} = \theta_0 + \Delta\theta \frac{t}{T} \tag{2}$$

The feedforward action is obtained by mapping the reference angle to -1~1:

$$a_{ff} = 2\frac{\theta_t^{ref} - \theta_{\min}}{\theta_{\max} - \theta_{\min}} - 1 \tag{3}$$

where θ_t^{ref} is the reference angle of the joint, and $\theta_{\min}, \theta_{\max}$ represent the lower limit and upper limit of the joint range of motion, respectively.

For legged locomotion, an important feature is the foot swings up and down periodically. Therefore, we decided to use the foot swing motion as feedforward. For legged robots, by using different foot swing sequences and swing trajectories, various gaits can be easily learned. In this paper, we explore various gaits including bipedal walking, level walking, march walking, hopping, and jumping, as well as quadrupedal trotting, pacing, bounding, and pronking. The detailed feedforward action for each gait can be found in the attached video. For most gaits, we only use the sinusoidal trajectory (1) to generate the feedforward. Only for biped march walking and biped hopping, we have applied a combination of the sinusoidal trajectory (1) and the ramp trajectory (2) to design the feedforward.

C. *Learning Setup*

- Hyperparameter

We use PPO for reinforcement learning. The hyperparameters are: batch_size 2048, buffer_size 20480, learning_rate 0.0003, beta 0.005, epsilon 0.2, lambda 0.95, num_epoch 3.

- Neural network

The neural network has an actor-critic structure, where the actor network is a multi-layer perceptron (MLP) with 3 hidden layers with 512 hidden units for each layer and the critic network is an MLP with 2 hidden layers with 128 hidden units for each layer. The output of the actor network is clipped to -1~1.

- Observation

The observations are

$$o_t = [q_t \quad \omega_t \quad v_t \quad \theta_t \quad \dot{\theta}_t]^T \tag{4}$$

When RO is applied, the observations are

$$o_t = [q_t \quad \omega_t \quad v_t \quad \theta_t \quad \dot{\theta}_t \quad \theta_t^{ref}]^T \tag{5}$$

where q_t is the orientation of the body expressed by a quaternion, containing 4 variables. ω_t, v_t are the body angular velocity and body linear velocity in the body frame, respectively, each containing 3 variables. $\theta_t, \dot{\theta}_t, \theta_t^{ref}$ are the joint angle, joint angular velocity, and joint reference angle, respectively, each containing 12 variables (for both biped and quadruped robots).

- Action

To enhance the smoothness of the action, the output of the actor network is passed through a low-pass filter to obtain the feedback action. Denote the original output of the actor network as a_{nn}, then the feedback action is obtained by

$$a_{fb} = 0.9a_{fb}^{last} + 0.1a_{nn} \tag{6}$$

where a_{fb}^{last} is the value of a_{fb} in the last time step. Then the feedback action a_{fb} is weighted by k_b and added with the feedforward to obtain the control action

$$a_t = a_{ff} + k_b a_{fb} \tag{7}$$

The control action a_t should be within the range -1~1. Since a_{ff}, a_{fb} are both within -1~1, we choose k_b within 0~2. However, the obtained a_t may be out of the range -1~1. Therefore, we clip it to calculate the command joint angle

每条腿只有四个关节。为了增强其能力，我们在仿真模型中增加了髋关节偏航和踝关节翻滚关节。对于四足机器人，我们使用Unitree A1模型。两款机器人的规格如下。

双足机器人：每条腿有6个主动关节。各关节的运动范围（以图4位置为零点）为（单位：度）：髋关节偏航 -30~30，髋关节翻滚 -30~30，髋关节俯仰 -60~120，膝关节俯仰 -170~10，踝关节翻滚 -30~30，踝关节俯仰 -60~60。最大关节力为200牛米。关节位置控制回路的刚度为2000，阻尼为100。

四足机器人：每条腿有3个主动关节。各关节的运动范围（以图4位置为零点）为（单位：度）：髋关节翻滚 -46~46，髋关节俯仰 -90~210，膝关节俯仰 -94.5~7.5。最大关节力为33.5牛米。关节位置控制回路的刚度为180，阻尼为8。

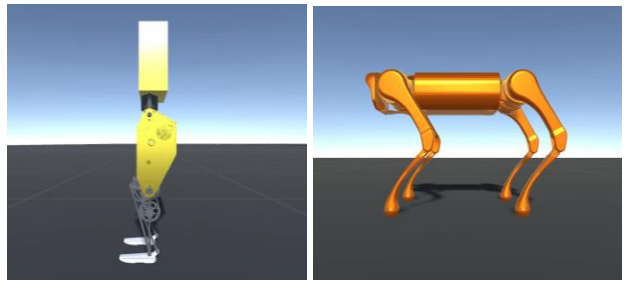


图4. 模拟的机器人模型。左：Ranger Max；右：Unitree A1。

B. 前馈设计

前馈是每个关节的位置参考轨迹。为简单起见，我们考虑两种常用的轨迹。

第一种是周期轨迹，其中关节以固定周期循环运动。在这种情况下，参考角度被设计为正弦信号

$$\theta_t^{ref} = \theta_0 + \Delta\theta \frac{1-\cos\left(2\pi t / T\right)}{2} \tag{1}$$

第二种是关节从一个位置移动到另一个位置的轨迹。在这种情况下，参考角度被设计为斜坡信号

$$\theta_t^{ref} = \theta_0 + \Delta\theta \frac{t}{T} \tag{2}$$

前馈通过将参考角度映射到 -1~1 来实现：

$$a_{ff} = 2\frac{\theta_t^{ref} - \theta_{\min}}{\theta_{\max} - \theta_{\min}} - 1 \tag{3}$$

其中 θ_t^{ref} 是关节的参考角度，$\theta_{\min}, \theta_{\max}$ 分别代表关节运动范围的下限和上限。

对于腿部运动而言，一个重要的特征是足部会周期性地上下摆动。因此，我们决定将足部摆动运动用作前馈。对于腿式机器人，通过使用不同的足部摆动序列和摆动轨迹，可以轻松学习各种步态。在本文中，我们探讨多种步态，包括双足行走、平地行走、正步走、行走、单脚跳和跳跃，以及四足小跑、踱步、弹跳和腾跃。每种步态的详细前馈动作可在附带的视频中找到。对于大多数步态，我们仅使用正弦轨迹（1）来生成前馈。仅针对双足行军行走和双足单脚跳，我们采用了正弦轨迹（1）与斜坡轨迹（2）的组合来设计前馈。

C. 学习设置

- 超参数

我们使用PPO进行强化学习。超参数为：批量大小2048，缓冲区大小20480，学习率0.0003，beta 0.005，epsilon 0.2，lambda 0.95，训练轮数3。

- 神经网络

该神经网络采用演员-评论家结构，其中演员网络是一个多层感知机（MLP），包含3个隐藏层，每层512个隐藏单元；评论家网络是一个MLP，包含2个隐藏层，每层128个隐藏单元。演员网络的输出被裁剪至 -1~1。

- 观测

观测值为

$$o_t = [q_t \quad \omega_t \quad v_t \quad \theta_t \quad \dot{\theta}_t]^T \tag{4}$$

当应用RO时，观测值为

$$o_t = [q_t \quad \omega_t \quad v_t \quad \theta_t \quad \dot{\theta}_t \quad \theta_t^{ref}]^T \tag{5}$$

其中 q_t 由四元数表示，包含4个变量。ω_t, v_t 分别为机体坐标系下的机体角速度和机体线速度，各包含3个变量。$\theta_t, \dot{\theta}_t, \theta_t^{ref}$ 分别为关节角度、关节角速度和关节参考角度，各包含12个变量（适用于双足和四足机器人）。

- 动作

为增强动作的平滑性，将演员网络的输出通过一个低通滤波器，得到反馈动作。记演员网络的原始输出为 a_{nn}，则反馈动作由下式得到

$$a_{fb} = 0.9a_{fb}^{last} + 0.1a_{nn} \tag{6}$$

其中 a_{fb}^{last} 是 a_{fb} 在上一时间步的值。然后，反馈动作 a_{fb} 以 k_b 为权重加权，并与前馈相加，得到控制动作

$$a_t = a_{ff} + k_b a_{fb} \tag{7}$$

控制动作 a_t 应处于范围 -1~1 内。

由于 a_{ff}，a_{fb} 均位于 -1~1 范围内，我们选择 k_b 在 0~2 内。然而，所获得的值 a_t 可能超出 -1~1 的范围。因此，我们对其进行裁剪以计算指令关节角度

$$\theta_t^{cmd} = \theta_{\min} + \frac{\mathrm{sat}(a_t)+1}{2}(\theta_{\max}-\theta_{\min}) \tag{8}$$

where sat() represents the linear saturation function. The command angle θ_t^{cmd} is sent to the position control loop of each joint.

D. Reward Design

Due to the application of the feedforward, there is no need for a tedious reward design. Indeed, we found a simple reward is enough to generate various natural-looking gaits with instruction learning. We design the reward as a combination of several components. Each component is explained as follows

1) The mimic reward r_m is designed as $r_m = 1.2 - 0.02\sum_1^{12}\left|\left(\theta_i - \theta_i^{ref}\right)\right|$ (angle unit: degree), which encourages the joint to track the reference trajectory.

2) The alive reward r_a is designed as $r_a = 1$. The robot receives this reward per time step for not falling.

3) The balance reward r_b is designed as $r_b = -0.1\left(\left|\theta_{pitch}\right| + \left|\theta_{yaw}\right| + \left|\theta_{roll}\right|\right)$ (angle unit: degree), which encourages keeping the body upright.

4) For the velocity reward, we give three options

- $r_v^{step} = -\left|v_{forward}\right| - \left|v_{lateral}\right| - \left|v_{vertical}\right|$
- $r_v^{walk} = v_{forward} - \left|v_{lateral}\right| - \left|v_{vertical}\right|$
- $r_v^{jump} = v_{forward} - \left|v_{lateral}\right| + \left|v_{vertical}\right|$

where the first encourages the robot to keep its body still; the second encourages moving forward; and the third encourages moving forward and meanwhile moving up and down.

5) The synchronization reward r_s is designed as $r_s = -0.05\sum_1^{3}\left|\left(\theta_i^{left} - \theta_i^{right}\right)\right|$ (angle unit: degree), which encourages the biped robot to synchronize the motion of the two legs (θ_i represents pitch angle of hip, knee, and ankle).

A combination of the rewards above can be used to generate different gaits. Table I lists the rewards we applied for some commonly seen gaits.

Table 1. Reward design.

Gaits	Reward
Stepping (quadruped and biped)	$r = r_m + r_a + r_b + r_v^{step}$
Biped: walk, level walk, march walk Quadruped: trot, pace, bound	$r = r_a + r_b + r_v^{walk}$
Biped: hop Quadruped: pronk	$r = r_a + r_b + r_v^{jump}$
Biped: jump	$r = r_a + r_b + r_v^{jump} + r_s$

Remark: It can be seen from Table I that only the stepping motion has adopted a mimic reward. Instruction learning actually does not require a mimic reward to learn a motor skill. For the stepping motion, we add the mimic reward just to make a fair comparison with imitation learning. Removing the mimic reward will not affect instruction learning to learn the stepping motion.

Termination condition: we end the episode if the robot “falls” ($\left|\theta_{pitch}\right| > 10\deg$ or $\left|\theta_{roll}\right| > 10\deg$) or if the time step reaches 1000.

V. Simulation Results

We use Unity ML-Agents for simulation, which adopts the Physix engine. The time step for each action is 0.01 seconds, which indicates a control frequency of 100 Hz. The training is running on the CPU. To speed up the training, we use 12 copies of the agents for parallel training. In our experience, it generally takes about 10 seconds to complete 10k training steps.

A. Stepping Motion Learning

We first test the stepping motion learning. The aim is to let the robot learn to step in place while maintaining balance. The feedforward is a periodic stepping motion (each foot goes up and down when the robot is hanging up but it will fall when put on the ground).

- **Instruction Learning vs. Imitation Learning**

We compare four learning methods: imitation learning (IML), imitation learning with reference in the observation (IML-RO), instruction learning (INL), and instruction learning with reference in the observation (INL-RO). For IML and IML-RO, the action is $a_t = a_{fb}$. For INL and INL-RO, the action is $a_t = 0.5a_{fb} + a_{ff}$. To make a fair comparison, we use the same reward (with a mimic reward, see Table I) for all of them. The reward curves are shown in Fig. 5 for the biped robot and Fig. 6 for the quadruped robot.

From the two figures, we have the following observations:

1) For instruction learning, the influence of RO can be neglected as can be seen that the learning curves of INL and INL-RO are almost the same. This might be because the reference already enters the controller through the feedforward, therefore it is no need to add the reference in the observation.

2) RO is critical for imitation learning. Without RO, imitation learning can hardly learn the stepping motion as can be seen that the reward of IML converges to a much smaller value (the robot actually learns to stand still).

3) Instruction learning is much more efficient than imitation learning, even when RO is applied. Comparing the results of INL with IML-RO, it can be seen that during the initial learning process, INL spends much fewer time steps to achieve the same reward, especially for the quadruped robot. This indicates the high efficiency of instruction learning. Moreover, INL still maintains a higher reward even at the post-learning stage.

$$\theta_t^{cmd} = \theta_{\min} + \frac{\mathrm{sat}(a_t)+1}{2}(\theta_{\max}-\theta_{\min}) \tag{8}$$

其中 表示线性饱和函数。指令角度 被发送到每个关节的位置控制环。

D. 奖励设计

由于采用了前馈，无需繁琐的奖励设计。事实上，我们发现一个简单的奖励就足以通过指令学习生成各种自然的步态。我们将奖励设计为多个组件的组合。每个组件的说明如下：

1) 模仿奖励 被设计为（角度单位：度），用于鼓励关节跟踪参考轨迹。

2) 存活奖励 被设计为 r_a。 机器人每个时间步因未摔倒而获得此奖励。

3) 平衡奖励 设计为(角度单位： 度)，用于鼓励保持身体直立。

4) 对于速度奖励，我们提供三种选项

- $r_v^{step} = -\left|v_{forward}\right| - \left|v_{lateral}\right| - \left|v_{vertical}\right|$
- $r_v^{walk} = v_{forward} - \left|v_{lateral}\right| - \left|v_{vertical}\right|$
- $r_v^{jump} = v_{forward} - \left|v_{lateral}\right| + \left|v_{vertical}\right|$

其中第一种鼓励机器人保持身体静止；第二种鼓励向前移动；第三种鼓励向前移动的同时上下运动。

5) 同步奖励 设计为(角度单位：度)，用于鼓励双足机器人同步两条腿的运动（代表髋关节、膝关节和踝关节的俯仰角）

θ_i

上述奖励的组合可用于生成不同的步态。表I列出了我们针对一些常见步态所应用的奖励。

表1. 奖励设计。

步态	奖励
踏步（四足和双足）	$r = r_m + r_a + r_b + r_v^{step}$
双足：行走、平地行走、正步走 四足：小跑、踱步、跳跃	$r = r_a + r_b + r_v^{walk}$
双足：单脚跳 四足：蹦跳	$r = r_a + r_b + r_v^{jump}$
双足：跳跃	$r = r_a + r_b + r_v^{jump} + r_s$

备注：从表I可以看出，只有踏步运动采用了模仿奖励。指令学习实际上并不需要模仿奖励来学习运动技能。对于踏步运动，我们添加模仿奖励只是为了与模仿学习进行公平比较。移除模仿奖励不会影响指令学习对踏步运动的学习。

终止条件：如果机器人“摔倒”（ 或 ）或 时间步达到 1000，则结束该回合。 10度 $\left|\theta_{roll}\right| >$

V. S仿真结果

我们使用Unity ML-Agents进行仿真，该工具采用了Physix引擎。每个动作的时间步为0.01秒，对应控制频率为100 Hz。训练在CPU上运行。为加速训练，我们使用12个智能体副本进行并行训练。根据经验，完成1万步训练通常需要约10秒。

A. 踏步运动学习 我们首先测试踏步运

动学习。目标是让机器人学会在保持平衡的同时原地踏步。前馈是一个周期性踏步运动（当机器人悬空时，每只脚会上下移动，但放在地面上时会摔倒）。

- 指令学习与模仿学习对比

我们比较了四种学习方法：模仿学习 (IML)、带观测参考的模仿学习 (IML-RO)、指令学习 (INL) 以及带观测参考的指令学习 (INL-RO)。对于 IML 和 IML-RO，动作是 。 对于 INL 和 INL-RO，动作是 。为了公平比较，我们对所有方法使用相同的奖励（包含模仿奖励，见表I）。奖励曲线如图5（双足机器人）和图6（四足机器人）所示。

从这两张图中，我们得到以下观察结果：

1) 对于指令学习，RO 的影响可以忽略不计，因为可以看到 INL 和 INL-RO 的学习曲线几乎相同。这可能是因为参考已经通过前馈进入了控制器，因此无需在观测中添加参考。

2) RO 对模仿学习至关重要。没有 RO，模仿学习几乎无法学习踏步运动，可以看到 IML 的奖励收敛到了一个更小的值（机器人实际上学会了静止站立）。

3) 即使应用了RO，指令学习也比模仿学习高效得多。对比INL与IML-RO的结果可以看出，在初始学习过程中，INL花费更少的时间步就能达到相同的奖励，尤其是对于四足机器人而言。这表明了指令学习的高效性。此外，即使在后期学习阶段，INL仍能保持更高的奖励。

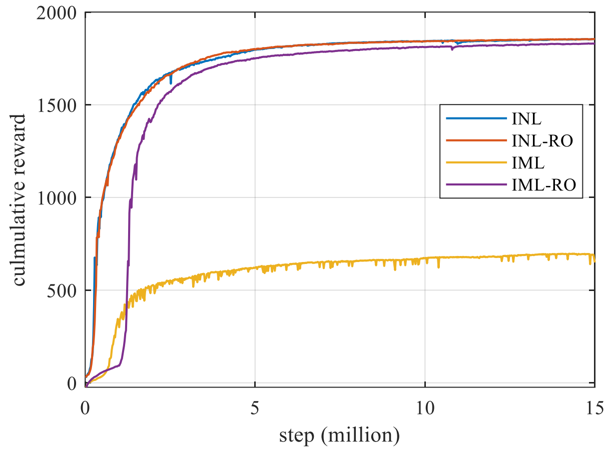


Figure 5. Reward curves for biped stepping.

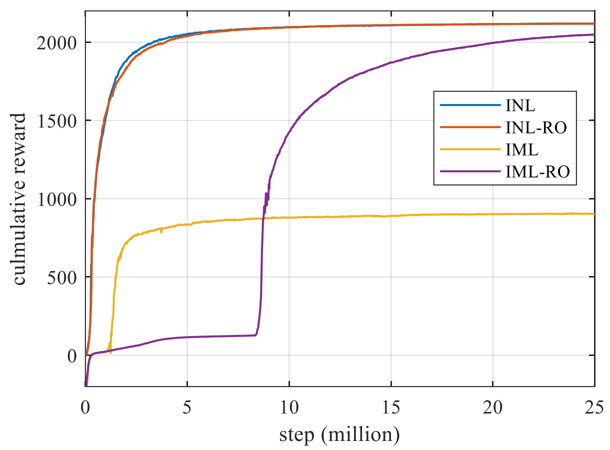


Figure 6. Reward curves for quadruped stepping.

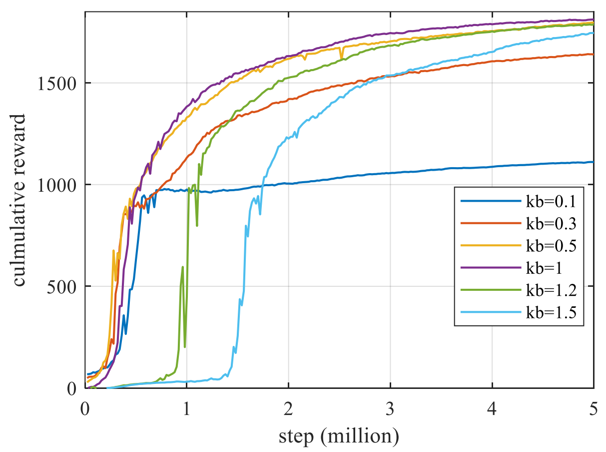


Figure 7. Reward curves for biped stepping with different k_b.

- **Influence of the action bounding range**

To investigate the influence of the action bounding range, we take the biped robot as an example and select different k_b values. The learning results are shown in Fig. 7. It can be seen that the action bounding range has a significant influence on the learning performance. If the range is too big (see $k_b = 1.2$ and $k_b = 1.5$), then the guiding role of the feedforward reduces a lot and the learning notably slows down. If the range is too small (see $k_b = 0.1$ and $k_b = 0.3$), the learning is fast in the beginning but converges to a lower reward. This phenomenon is because the action range is too small to ensure good reference tracking, which ends up in a small mimic reward. When the range is properly selected (see $k_b = 0.5$ and $k_b = 1$), the learning is fast and the reward converges to a high value.

B. Locomotion Learning

In the previous stepping motion learning, we adopted a mimic reward. The function of the mimic reward is to encourage the robot to follow a reference trajectory. For imitation learning, the mimic reward is indispensable and crucial to learning the reference motion correctly. However, for instruction learning, the reference motion is directly used in the feedforward action, so there is no need for the mimic reward. In fact, there are many benefits of removing the mimic reward. Firstly, it is difficult to track a trajectory accurately, which is time-consuming for training and decreases learning efficiency. Secondly, the mimic reward constrains the motion. Removal of the mimic reward can better release the robot's potential to achieve other goals rather than just following the reference.

Therefore, for the next locomotion learning tasks, we no longer use the mimic reward or RO. With the same feedforward as the stepping motion, locomotion can be easily achieved by simply altering the velocity reward (see Table I for details). As a result, the biped robot learns to walk and the quadruped robot learns to trot. Moreover, by reorganizing the feedforward signal, the robots can easily learn other gaits. For the biped robot, we demonstrate learning of walking, level walking, march walking, jumping, and hopping. And for the quadruped robot, we demonstrate learning of trotting, pacing, bounding, and pronking. Fig. 8 and Fig. 9 show the reward curves, where we only train for 1 million steps for the biped robot and 3 million steps for the quadruped robot, which are enough to learn all the desired gaits. This indicates that instruction learning is highly efficient. Snapshots of the resulting gaits are shown in Fig. 10 and Fig. 11, respectively.

The results also verify the versatility of instruction learning. By using different feedforward signals, the robots can learn diverse locomotion tasks with simple (and even the same) rewards. Besides, compared to imitation learning, the reference signal in instruction learning can be much simpler, it only roughly characterizes the motion style and does not need to be very exact, therefore it can be manually designed instead of using motion capture data.

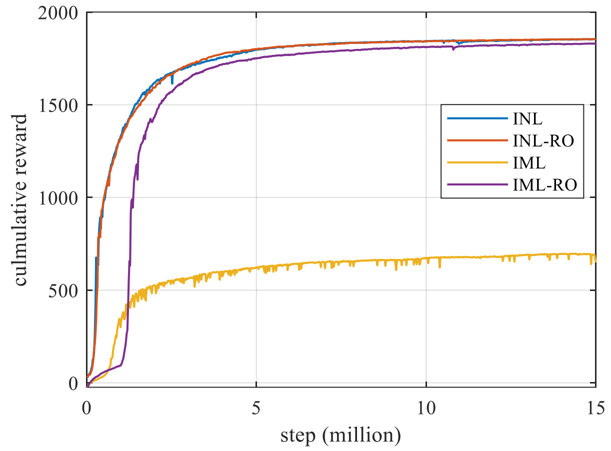


图5. 双足行走的奖励曲线。

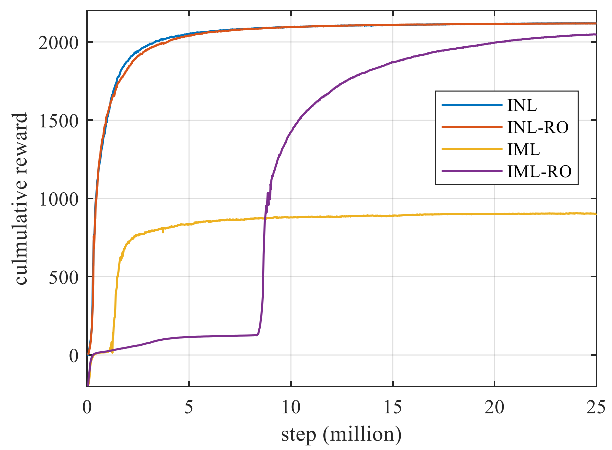


图6. 四足行走的奖励曲线。

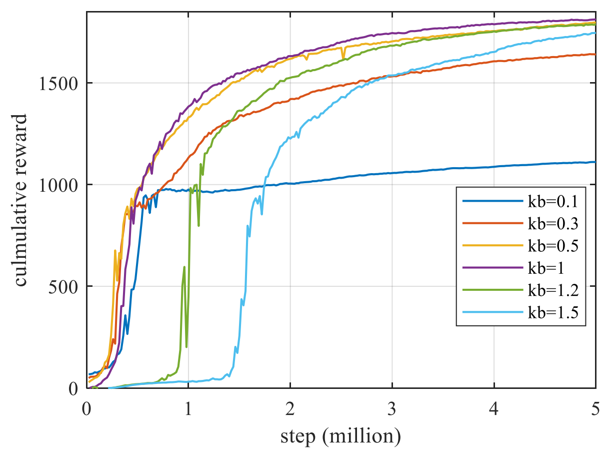


图7. 不同条件下的双足行走奖励曲线 k_b.

- 动作边界范围的影响

为了探究动作边界范围的影响，我们以双足机器人为例，选取不同的数值。学习结果如图7所示。可以看出，动作边界范围对学习性能有显著影响。如果范围过大（参见 $k_b = 1.2$ 和），则前馈的引导作用会大幅降低，学习速度明显减慢。如果范围过小（参见 和），则学习初期速度较快，但最终收敛到的奖励值较低。这种现象是因为动作范围过小，无法保证良好的参考跟踪，从而导致模仿奖励较小。当范围选择适当时（参见 和），学习速度快，且奖励收敛到较高值。

B. 运动学习

在之前的踏步运动学习中，我们采用了模仿奖励。模仿奖励的功能是鼓励机器人跟随参考轨迹。对于模仿学习而言，模仿奖励对于正确学习参考动作是不可或缺且至关重要的。然而，对于指令学习，参考动作直接用于前馈动作中，因此无需模仿奖励。事实上，移除模仿奖励有许多好处。首先，精确跟踪轨迹很困难，这会耗费训练时间并降低学习效率。其次，模仿奖励会约束动作。移除模仿奖励可以更好地释放机器人的潜力，使其能够实现其他目标，而不仅仅是跟随参考。

因此，在接下来的运动学习任务中，我们不再使用模仿奖励或RO。采用与踏步运动相同的前馈，只需改变速度奖励（详见 表I）即可轻松实现运动。结果，双足机器人学会了行走，四足机器人学会了小跑。此外，通过重新组织前馈信号，机器人可以轻松学习其他步态。对于双足机器人，我们展示了行走、平地行走、行军行走、跳跃和单脚跳的学习。对于四足机器人，我们展示了小跑、踱步、弹跳和腾跃的学习。图8和图9展示了奖励曲线，我们仅对双足机器人训练了100万步，对四足机器人训练了300万步，这足以学习所有期望的步态。这表明指令学习效率极高。最终步态的截图分别如图10和图11所示。

结果也验证了指令学习的通用性。通过使用不同的前馈信号，机器人能够以简单（甚至相同）的奖励学习多样化的运动任务。此外，与模仿学习相比，指令学习中的参考信号可以简单得多，它们仅粗略表征运动风格，无需非常精确，因此可以手动设计，而无需使用动作捕捉数据。

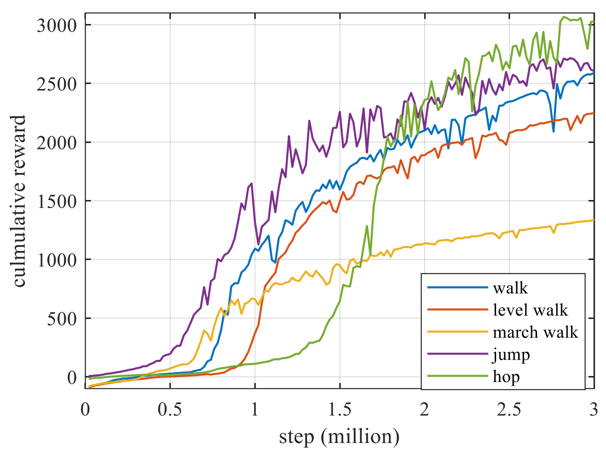


Figure 8. Reward curves for biped locomotion.

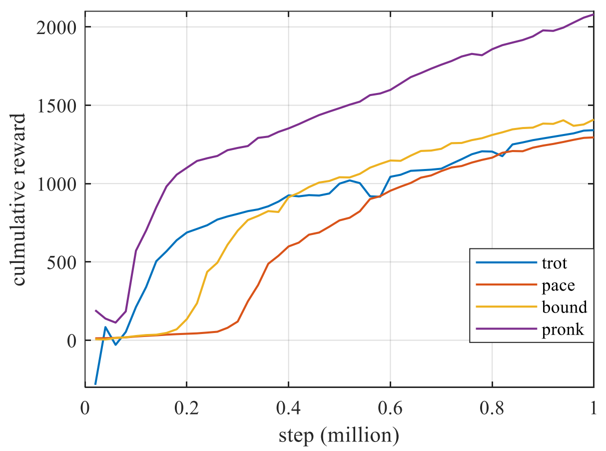
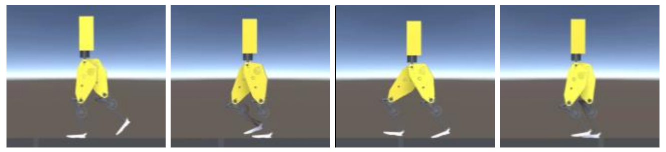
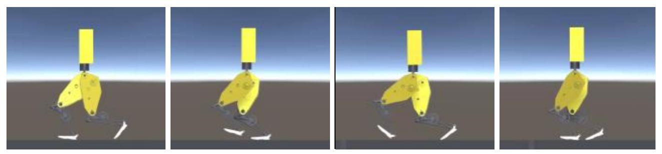


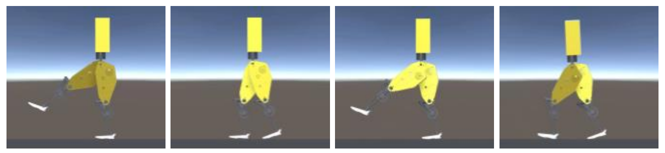
Figure 9. Reward curves for quadruped locomotion.



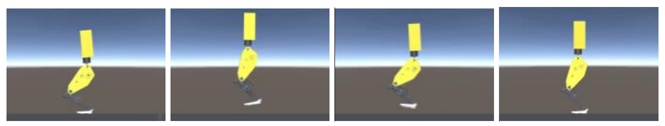
(A) Walk



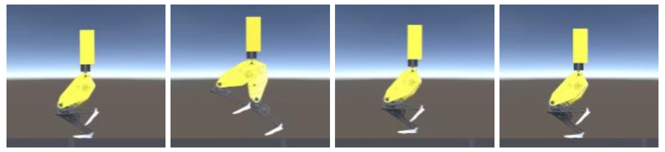
(B) Level walk



(C) March walk

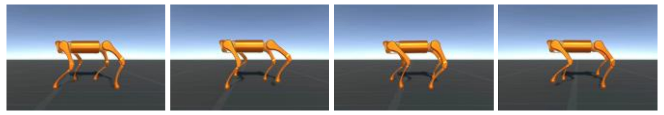


(D) Jump

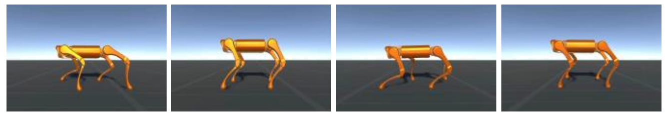


(E) Hop

Figure 10. Biped learned locomotion behavior.

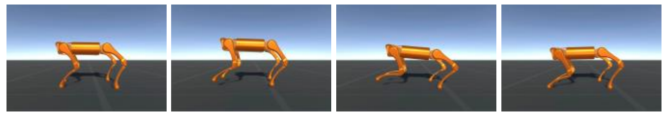


(A) Trot



(B) Pace

(C) Bound



(D) Pronk

Figure 11. Quadruped learned locomotion behavior.

C. Motion Adaptation

Despite the superior learning efficiency, another advantage of instruction learning is its flexibility. For traditional reinforcement learning, since the neural network weights cannot be manually tuned, if we want to adjust the learned motion, the only way is to retrain the neural network. But for instruction learning, there is a feedforward that can be directly adjusted after training. Therefore, we can adjust the learned motion conveniently. As we tested, the learned motion can be maintained when the feedforward is modified in a certain range.

To illustrate this, we use the period and amplitude of the feedforward (corresponds to step period and step height for walking) as examples. When training is completed for a gait, we change the period or amplitude (independently) of the feedforward bit by bit until the gait cannot be maintained (the robot falls or does not move forward), and then we record the adaption range, which is shown in Fig. 12. It can be seen that for all gaits, a feasible range exists. The adaption range can even be as low as 0.2 and as high as 4. For some gaits like biped march walking, jumping, and hopping, the adaption range is smaller compared to other gaits, this might be because those gaits are more unstable.

The motion adaption property of instruction learning is very useful, allowing us to directly adjust the learned motion and thus eliminate the retraining time. The adaption capability is also similar to human skill learning. Once we learn a motor skill, we can easily master the variations of it. For example, we can change the gait frequency and gait clearance height

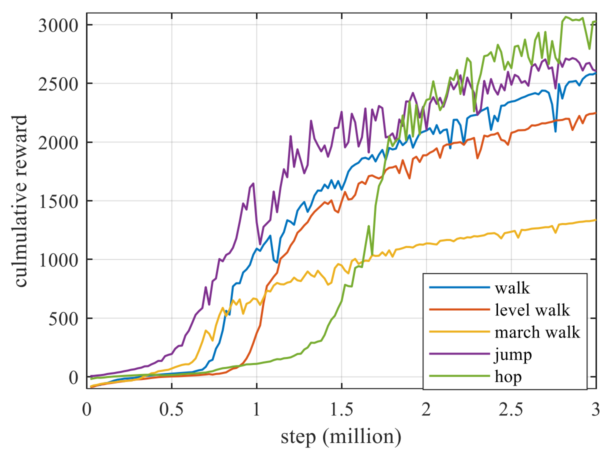


图8. 双足运动的奖励曲线。

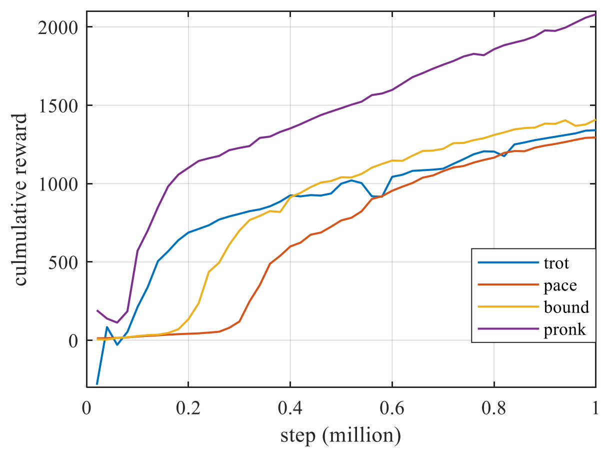
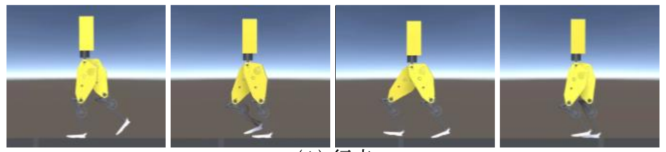
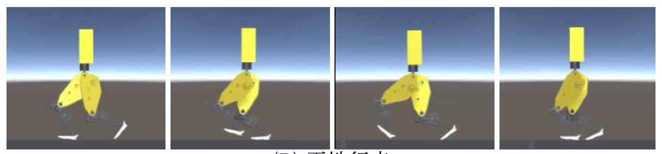


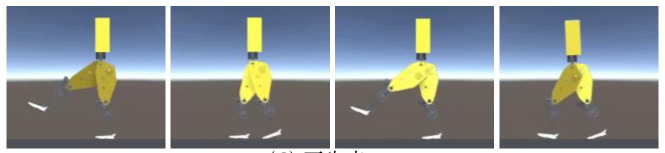
图9. 四足运动的奖励曲线。



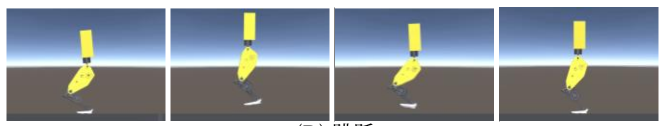
(A) 行走



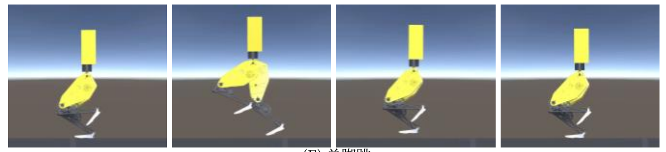
(B) 平地行走



(C) 正步走

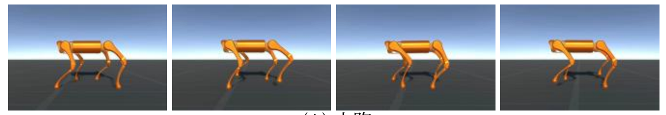


(D) 跳跃

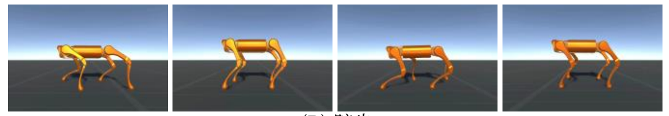


(E) 单脚跳

图10. 双足机器人习得的运动行为。

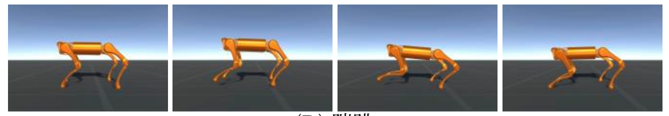


(A) 小跑



(B) 踱步

(C) 跳跃



(D) 蹦跳

图11. 四足机器人习得的运动行为。

C. 运动适应

除了卓越的学习效率外，指令学习的另一个优势在于其灵活性。对于传统强化学习而言，由于神经网络权重无法手动调整，若要调整已习得的运动，唯一方法是重新训练神经网络。但指令学习则不同，其前馈信号可在训练后直接调整，因此我们可以便捷地调整已习得的运动。经测试，当在一定范围内修改前馈信号时，已习得的运动仍能保持稳定。

为了说明这一点，我们以前馈的周期和振幅（对应行走时的步态周期和步高）为例。当某个步态的训练完成后，我们逐步改变前馈的周期或振幅（独立进行），直到步态无法维持（机器人摔倒或无法前进），然后记录适应范围，如图12所示。可以看出，对于所有步态，都存在一个可行范围。适应范围最低可至0.2，最高可达4。对于某些步态，如双足行军行走、跳跃和单脚跳，其适应范围相比其他步态更小，这可能是因为这些步态更不稳定。

指令学习的运动适应特性非常实用，使我们能够直接调整已习得的运动，从而省去重新训练的时间。这种适应能力也与人类技能学习相似。一旦我们掌握了一项运动技能，就能轻松掌握其变体。例如，我们可以改变步态频率和步态离地高度。

easily without relearning.

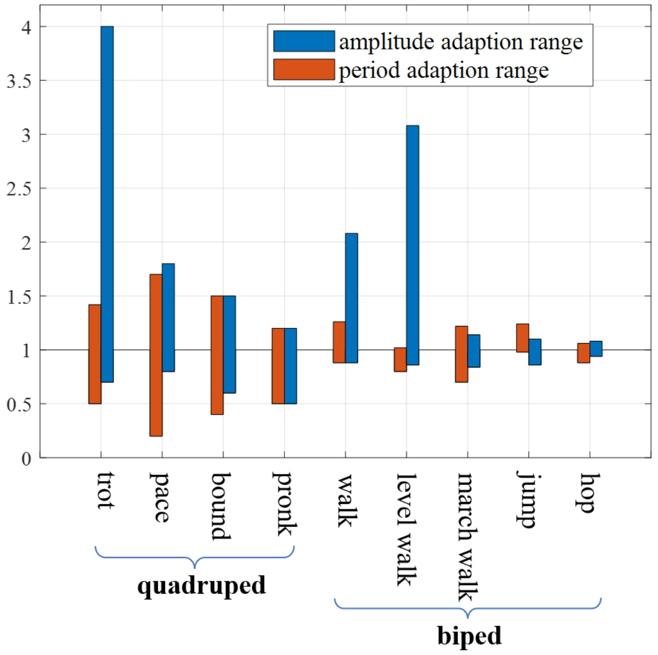


Figure 12. Adaption range of learned motion to a modified feedforward.

VI. Experimental Results

To verify the effectiveness of instruction learning for real systems, we do sim-to-real transfer and online learning experiments on real quadruped robots. Compared to the previous simulations, a slight modification is made to the observation. First, we remove body velocity from the observation since it is not so accurate. Second, the body orientation quaternion in the observation is replaced by the body pitch and roll angles to ensure consistency when the robot faces different directions.

A. *Sim-to-Real Transfer*

We have two groups working on the sim-to-real transfer. One group uses the Raisim software and the Unitree A1 robot, while the other group uses the IsaacGym software and the Unitree Go1 robot.

Figure 13. A1 robot at different gaits. Left to right: trot, pace, bound, pronk.

Figure 14. GO1 robot robust walking.

The first group focuses on realizing the four categories of gaits. We first try to directly transfer the learned policy to the real robot but the robot cannot keep balance due to the sim-to-real gap. We then add random disturbances to the robot during training, which results in a successful transfer, but the motion is not so robust (see Fig. 13 and the video). The second group focuses on command following and robust walking. We add the body forward velocity command and yaw velocity command in the observation and the reward and apply domain randomization during training. As a result, the robot can follow a given velocity command and maintain balance when disturbed (see Fig. 14 and the video). It can also adapt to the feedforward modification in a certain range.

B. *Online Learning*

Due to the high learning efficiency, instruction learning is very well suited for real robot online learning. Online learning collects data from the real robot and updates the control policy in real time, which does not have the sim-to-real problem. However, compared to learning in simulation, online learning cannot be accelerated or trained parallelly, therefore it particularly requires highly efficient learning. Besides, resetting is troublesome in online learning when the robot falls. Instruction learning can handle these two problems thanks to its high efficiency and reduced action space which can prevent the robot from falling.

Recall the previous simulation results in Fig. 9 where the quadruped robots can learn various gaits in one million steps (time step 0.01s), which equals 2.78 hours in real time. That is about the battery life on one charge and is acceptable. However, the robot will easily fall with the feedback ratio of 0.5. Besides, as the robot walks faster and faster, we need a big enough field for the experiment. Therefore, we decided to use a smaller feedback ratio and train for a shorter time.

We select the trotting gait as an example. The robot starts from a stepping-in-place motion and learns to turn or walk forward. Before each experiment, we try a more realistic simulation first to ensure the feasibility. The reset is removed in the simulation, and the reward only uses available data the same as the real robot. When the simulation is passed, we then apply online learning to the real robot. The experimental scene is shown in Fig. 15.

During training, the robot learns on intermittent episodes. Each episode takes about 5.5 s and includes three stages: 1) The robot walks for about 3.3 s; 2) PPO updates; 3) The robot takes several steps to recover to the initial state. Both online learning tasks take about 20 minutes for training. The robot does not fall during the training and we only need to reposition it when necessary. The reward curves for the two online learning experiments are shown in Fig. 16.

Figure 15. Online learning scene.

无需重新学习即可轻松实现。

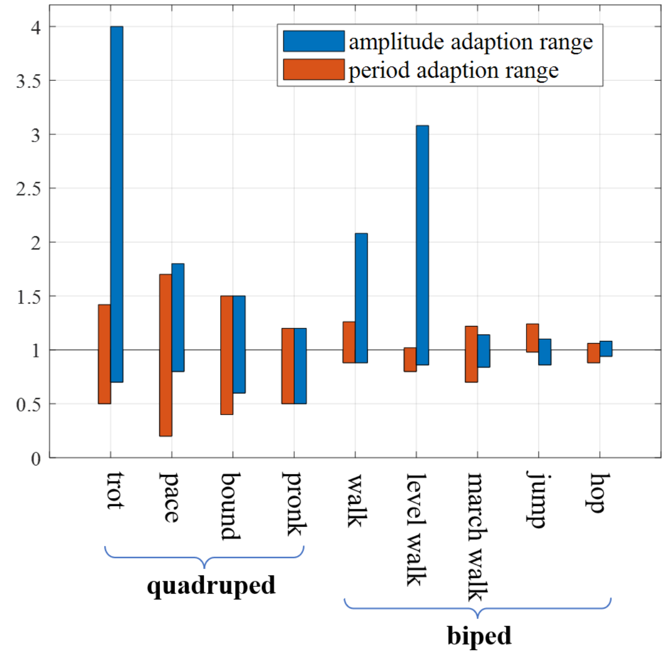


图12. 学习到的运动对修改后前馈的适应范围。

VI. 实验结果

为了验证指令学习在实际系统中的有效性，我们在真实四足机器人上进行了仿真到现实迁移和在线学习实验。与之前的仿真相比，我们对观测做了一些细微调整。首先，我们从观测中移除了身体速度，因为它不够精确。其次，观测中的身体朝向四元数被替换为身体俯仰角和横滚角，以确保机器人在面对不同方向时的一致性。

A. 仿真到现实迁移

我们有两个小组负责仿真到现实迁移工作。一组使用Raisim软件和宇树A1机器人，另一组使用IsaacGym软件和宇树Go1机器人。

图13. A1机器人不同步态展示。从左至右：小跑、踱步、跳跃、蹦跳。

图14. GO1机器人鲁棒行走。

第一组专注于实现四类步态。我们首先尝试将学习到的策略直接迁移到真实机器人上，但由于仿真到现实差距，机器人无法保持平衡。随后，我们在训练过程中向机器人添加随机扰动，这实现了成功迁移，但运动不够鲁棒（见图13及视频）。第二组专注于指令跟随与鲁棒行走。我们在观测和奖励中加入了身体前向速度指令和偏航速度指令，并在训练中应用域随机化。最终，机器人能够跟随给定的速度指令，并在受到扰动时保持平衡（见图14及视频）。它还能在一定范围内适应前馈修改。

B. 在线学习

由于学习效率高，指令学习非常适合真实机器人的在线学习。在线学习从真实机器人收集数据并实时更新控制策略，因此不存在仿真到现实的问题。然而，与在仿真中学习相比，在线学习无法加速或并行训练，因此特别需要高效的学习。此外，当机器人摔倒时，在线学习中的重置操作也很麻烦。指令学习凭借其高效率和缩小的动作空间（可防止机器人摔倒），能够处理这两个问题。

回顾之前图9中的仿真结果，四足机器人可以在一百万步（时间步长0.01秒）内学习各种步态，这相当于现实时间中的2.78小时。这大约相当于一次充电的电池续航时间，是可以接受的。然而，当反馈比率为0.5时，机器人很容易摔倒。此外，随着机器人行走速度越来越快，我们需要足够大的场地来进行实验。因此，我们决定使用较小的反馈比率并缩短训练时间。

我们以小跑步态为例。机器人从原地踏步运动开始，学习转向或向前行走。每次实验前，我们首先尝试更逼真的仿真以确保可行性。仿真中取消了重置，奖励仅使用与真实机器人相同的可用数据。仿真通过后，我们将在线学习应用于真实机器人。实验场景如图15所示。

在训练过程中，机器人通过间歇性回合进行学习。每个回合大约持续5.5秒，包含三个阶段：1) 机器人行走约3.3秒；2) PPO更新；3) 机器人执行若干步以恢复至初始状态。两项在线学习任务各需约20分钟完成训练。训练期间机器人不会摔倒，仅在必要时需重新调整其位置。两次在线学习实验的奖励曲线如图16所示。

图15. 在线学习场景。

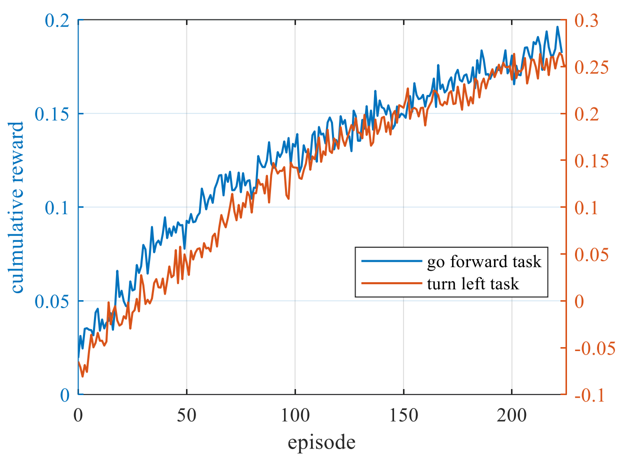


Figure 16. Reward curves for online learning.

For the turning task, the yaw velocity of the robot which can be obtained from the IMU is used as the reward. To guarantee fast learning, we apply a feedback ratio of 0.15 for the hip roll joints and 0 for the other joints. As can be seen from the video, the robot has a turning right trend in the beginning but successfully learns to turn left in the end.

For the walking forward task, the feedback ratio is 0 for the hip roll joints and 0.07 for the other joints. A problem in this task is that the reward requires to know the linear velocity of the robot but it cannot be obtained directly. We have tried to use a Kalman filter to estimate the velocity, which fuses the acceleration reading, the encoder reading, and the contact information, but the estimation performance is bad (especially when the velocity is small), which degrades the learning performance. Therefore, we give up using the velocity in the reward but instead use the angular velocity of the leg. We denote the leg angle as $\theta_{leg} = \theta_{hfe} + \theta_{kfe} / 2$ ($\theta_{hfe}, \theta_{kfe}$ represent the hip pitch and knee pitch angle, respectively), which is the pitch angle between the robot body and the virtual leg connecting the foot and the hip. Then the leg angular velocity is calculated as the leg angle difference between two time steps $\omega_{leg} = (\theta_{leg})_{k+1} - (\theta_{leg})_k$. Then the reward is designed as $r = \omega_{sp_1} + \omega_{sp_2} - \omega_{sw_1} - \omega_{sw_2}$, where $\omega_{sp_1}, \omega_{sp_2}$ represent angular velocity of the two supporting legs and $\omega_{sw_1}, \omega_{sw_2}$ the two swing legs, respectively. Using this reward, the robot successfully learns to walk forward in 20 minutes.

VII. Conclusion

This paper proposes a novel reinforcement learning framework named instruction learning for robot motor skill learning and verifies its effectiveness through a bunch of locomotion learning examples. The key components of instruction learning are the feedforward-feedback learning structure, the action bounding technique, and the mimic reward elimination idea, which distinguish it from imitation learning and are critical for achieving highly efficient and flexible learning. The feedforward and feedback form a knowledge and data dual-driven learning system. The feedforward is knowledge-driven, which can be modified immediately, and the feedback is data-driven, which is learned gradually. The training process of instruction learning is like human practicing, which is a process from knowing to doing. The action bounding technique constrains the action searching range in a reduced space along the feedforward action and makes instruction learning highly efficient. Besides, instruction learning uses a simple reward and removes the mimic reward. It tries to learn a motor skill by following instructions rather than following the reference trajectory accurately. This gives instruction learning high flexibility. Finally, instruction learning also shows good adaptivity, where the learned behavior can adapt to a group of modified feedforward.

Instruction learning has many similarities to human learning, which provides an interesting perspective for us to view the motor skill learning mechanism of humans. Instruction learning is easy to understand and implement, which provides a valuable framework for efficient motion learning and has great potential for robot learning in the real world. It should be noted that we did not pay special attention to improving the robustness of the controller, so the behaviors learned by instruction learning are currently not as stable as state-of-the-art learning controllers. However, instruction learning can be combined with other sim-to-real methods to enhance the robustness of the learned controllers, which is necessary for more complex real-world applications.

Acknowledgment

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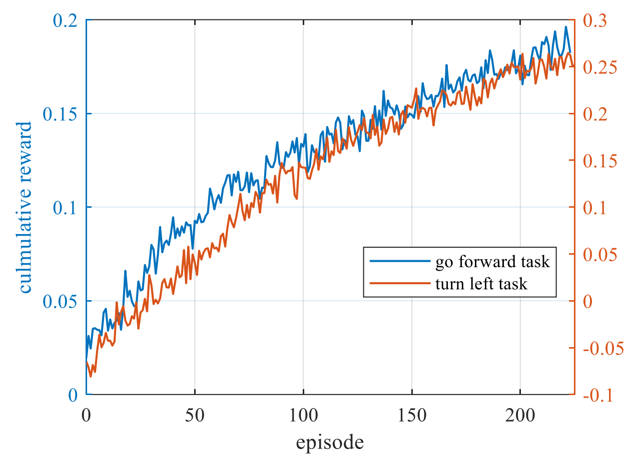

图16. 在线学习的奖励曲线。

对于转向任务，从惯性测量单元获取的机器人偏航角速度被用作奖励。为保证快速学习，我们对髋关节滚动关节应用0.15的反馈比率，对其他关节则应用0。从视频中可以看出，机器人起初有向右转的趋势，但最终成功学会了向左转。

对于向前行走任务，髋关节滚动关节的反馈比率为0，其他关节的反馈比率为0.07。该任务的一个问题是，奖励需要知道机器人的线速度，但无法直接获取。我们尝试使用卡尔曼滤波器来估计速度，该滤波器融合了加速度读数、编码器读数和接触信息，但估计性能较差（尤其是当速度较小时），这降低了学习性能。因此，我们放弃在奖励中使用速度，转而使用腿部的角速度。我们将腿部角度定义为 （ 分别表示髋关节俯仰角和膝关节俯仰角），即机器人身体与连接脚和髋关节的虚拟腿之间的俯仰角。然后，腿部角速度计算为两个时间步之间的腿部角度差。然后，奖励设计为，其中 表示两条支撑腿的角速度， 表示两条摆动腿的角速度。使用此奖励，机器人在20分钟内成功学会了向前行走。

$r = \omega_{sp_1} + \omega_{sp_2} - \omega_{sw_1} - \omega_{sw_2}$

$\omega_{sp_1}, \omega_{sp_2}$

$\omega_{sw_1}, \omega_{sw_2}$

VII. 结论

本文提出了一种名为指令学习的新型强化学习框架，用于机器人运动技能学习，并通过一系列运动学习实例验证了其有效性。指令学习的关键组成部分包括前馈-反馈学习结构、动作边界技术以及模仿奖励消除思想，这些特点使其区别于模仿学习，并且是实现高效灵活学习的关键。前馈与反馈构成了一个知识与数据双驱动学习系统。前馈是知识驱动的，可以立即修改；反馈是数据驱动的，需要逐步学习。指令学习的训练过程类似于人类练习，是一个从知道到做到的过程。动作边界技术将动作搜索范围约束在前馈动作周围的缩减空间中，从而使指令学习具有高效率。此外，指令学习使用简单的奖励并去除了模仿奖励。它试图通过遵循指令而非精确跟随参考轨迹来学习运动技能，这赋予了指令学习高度的灵活性。最后，指令学习还表现出良好的自适应性，即学习到的行为能够适应一组经过修改的前馈信号。

指令学习与人类学习有许多相似之处，这为我们观察人类的运动技能学习机制提供了一个有趣的视角。指令学习易于理解和实现，为高效的运动学习提供了宝贵的框架，并且在现实世界的机器人学习中具有巨大潜力。需要指出的是，我们并未特别关注提高控制器的鲁棒性，因此目前通过指令学习获得的行为不如最先进的学习控制器稳定。然而，指令学习可以与其他仿真到现实方法相结合，以增强所学控制器的鲁棒性，这对于更复杂的现实应用是必要的。

致谢

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